

VENOUS DISORDERS

Venous anatomy

- Veins = thin-walled (3 layers: adventitia, media, intima). **Endothelium = anticoagulant when intact.**
- **Valves** = bicuspid, in pairs; passive opening/closing with flow. **Rare/absent in pelvic veins + IVC.**
- **Leg veins:** deep (in muscle, all have valves), superficial (subcutaneous — **great + short saphenous**), perforating (~150, connect superficial→deep; mainly below knee).
- **Junctions: saphenofemoral + saphenopopliteal.**
- Normal flow = **one-way: superficial → deep → heart.**
- **Venous pump:** foot pump + **muscle (calf) pump** + valve system.

Chronic venous insufficiency / varicose veins

- **Valve failure → venous reflux.** Trendelenburg: varicose veins = malfunctioning valves.
- **Genetics = strong component** (hereditary; will get it regardless of risk factors if coded).
- **Myths corrected:** pelvic tumours/masses → **DVT not varicose veins** (compression → narrowing). **Pregnancy UNCOVERS hidden varicose veins** (oestrogen/progesterone dilate vessels) — doesn't cause them.
- **Chronic venous disease:** spectrum (spider veins → ulcers). Incidence 60–80%; **no gender difference.** Precipitants: prolonged standing/sitting, pregnancy, aging.
- **Spider veins = sign of underlying vein problem, NOT cosmetic.** Children/teens can have severe valve insufficiency ("snake legs").
- **Clinical:** achy/heavy legs, dragging, itching, foot/ankle oedema, cramps, burning, paraesthesia.
- **Gaiter area (lower leg) dermatological:** hyperpigmentation, eczema, lipodermatosclerosis, ulceration. **Leg ulcer in gaiter area = venous until proven otherwise.**
- **CEAP classification** (Clinical, Etiology, Anatomical, Pathophysiology).
- **Diagnosis: Doppler ultrasound** (gold standard for valve insufficiency).
- **Management:** curable (success 99.5%). Lifestyle (exercise, **compression socks**). **Endovenous techniques replaced surgery:** thermal (laser, radiofrequency ablation), non-thermal (foam sclerotherapy).

Deep Vein Thrombosis (DVT)

- Clot in deep veins (usually leg). Part of VTE — **3rd commonest CV death (after MI, stroke).**
- Usually starts in **calf (distal)**; extends to popliteal/femoral → symptomatic. **Untreated proximal DVT → 50% symptomatic PE.**
- **Complications:** PE, death, **post-thrombotic syndrome (20–50% of proximal symptomatic DVT — lifelong pain, swelling, ulcers).**
- **Classification:** location (proximal above knee / distal below / mixed), duration (acute ≤14d, subacute 14–28d, chronic >28d), aetiology (provoked vs unprovoked).
- **Risk factors:** **immobility** (stroke, surgery, hip/knee replacement, bed rest, flights), mechanical compression (tumour, pregnancy), vein injury (trauma, lines, drug abuse), **increased coagulation** — genetic (**protein C/S, antithrombin III deficiency, factor V Leiden**), acquired (cancer, sepsis, OCP, smoking), rheumatological (SLE, Behçet). Age ↑ risk; **gender not a risk factor.**

- **Pathophysiology: Virchow's triad** = vessel wall damage + turbulent flow + hypercoagulability.
- **Clinical:** pain, redness, swelling, limb oedema, tenderness, hot skin.
- **Evaluation: D-dimer (sensitive not specific)**, thrombophilia screen, **Doppler ultrasound**. **Wells score** (0–1 low, ≥ 2 high probability).
- **Treatment** (stop thrombus extension → prevent PE, ↓post-thrombotic syndrome):
- **Pharmacological: unfractionated heparin IV (initial/standard)**, LMWH (enoxaparin), DOACs (rivaroxaban, apixaban, fondaparinux).
- **Mechanical:** graduated compression socks, intermittent pneumatic compression, thrombectomy, iliac stenting, **IVC filters**.

High-yield one-liners

- **Valve failure → reflux → varicose veins; genetics strong; Doppler US = gold standard.**
- **Pregnancy uncovers (doesn't cause) varicose veins; pelvic masses → DVT not varicose.**
- **Gaiter area ulcer = venous until proven otherwise; spider veins = sign not cosmetic.**
- **Endovenous (laser/RFA/foam) replaced surgery.**
- **DVT: starts in calf; untreated proximal → 50% PE; post-thrombotic syndrome 20–50%.**
- **Virchow's triad; Wells score; D-dimer sensitive not specific; Doppler US confirms.**
- **DVT treatment: heparin (UFH initial) → LMWH/DOAC; IVC filter mechanical.**

VENOUS DISORDERS (Revision Table)

Bold = high-yield. Dr Dezhin Solai.

ANATOMY & CHRONIC VENOUS INSUFFICIENCY

Topic	Key points
Anatomy	Valves = bicuspid (absent in pelvic/IVC); deep (all valved), superficial (GSV , SSV), perforators (~150); SFJ + SPJ junctions ; one-way flow superficial→deep→heart; calf muscle pump
Pathophysiology	Valve failure → reflux; strong genetic component
Myths	Pelvic masses → DVT (not varicose); pregnancy UNCOVERS hidden varicose veins (hormonal)
Chronic venous disease	Spectrum (spider veins → ulcers); 60–80%; no gender diff; standing/sitting/aging
Clinical	Achy/heavy legs, oedema, itching; gaiter area: hyperpigmentation, eczema, lipodermatosclerosis, ulcer
Key facts	Gaiter ulcer = venous until proven otherwise; spider veins = sign not cosmetic; CEAP classification
Diagnosis/Mgmt	Doppler US = gold standard ; compression socks; endovenous (laser/RFA/foam) replaced surgery

DEEP VEIN THROMBOSIS

Topic	Key points
Overview	Clot in deep veins; 3rd CV death (after MI, stroke) ; starts in calf; untreated proximal → 50% PE
Complications	PE, death, post-thrombotic syndrome (20–50%)
Classification	Location (proximal/distal/mixed); duration (acute ≤14d, subacute 14–28d, chronic >28d); provoked vs unprovoked
Risk factors	Immobility (surgery, hip/knee replacement, flights), tumour/pregnancy, thrombophilia (protein C/S, AT3, factor V Leiden) , cancer, OCP, smoking
Pathophysiology	Virchow's triad: vessel damage + turbulent flow + hypercoagulability
Evaluation	D-dimer (sensitive not specific), Doppler US, Wells score (0–1 low, ≥2 high)
Treatment	UFH IV (initial/standard) → LMWH/DOAC (rivaroxaban); mechanical: compression, IVC filter